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Skill 28.1 Exercise 1

Refer to the movies DataFrame below.

	movie	production_budget	domestic_gross	worldwide_gross	mpaa_rating	genre
0	Evan Almighty	175000000.0	100289690.0	174131329.0	PG	Comedy
1	Waterworld	175000000.0	88246220.0	264246220.0	PG-13	Action
2	King Arthur: Legend of the Sword	175000000.0	39175066.0	139950708.0	PG-13	Adventure
3	47 Ronin	175000000.0	38362475.0	151716815.0	PG-13	Action
4	Jurassic World: Fallen Kingdom	170000000.0	416769345.0	1304866322.0	PG-13	Action

Suppose we want to compare the *production_budgets* for PG and R movies. Create two lists: one that has the *production_budget* for PG movies and another that has the *production_budget* for R movies.

Calculate the mean and median difference.

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Skill 28.2 Exercise 1

Refer to the movies DataFrame below.

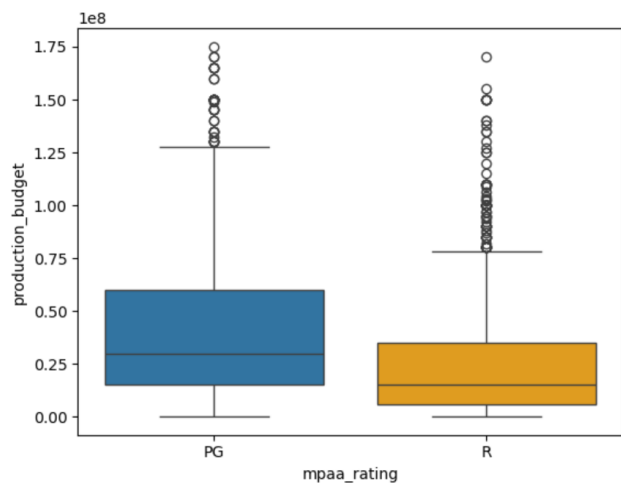
	movie	production_budget	domestic_gross	worldwide_gross	mpaa_rating	genre
0	Evan Almighty	175000000.0	100289690.0	174131329.0	PG	Comedy
1	Waterworld	175000000.0	88246220.0	264246220.0	PG-13	Action
2	King Arthur: Legend of the Sword	175000000.0	NaN	139950708.0	PG-13	Adventure
3	47 Ronin	NaN	38362475.0	151716815.0	PG-13	Action
4	Jurassic World: Fallen Kingdom	170000000.0	416769345.0	NaN	NaN	Action

Filter the dataframe for the movies with an *mpaa_rating* of R and PG only. Store the results in a new dataframe, *PG_R*

Complete the spaces below to make side by side box plots for R and PG movies.

```
sns.boxplot(data = _____,
             x = _____,
             y = _____,
             hue = _____,
             legend = _____)
plt.show()
```

Below are the actual box plots for PG and R movies,



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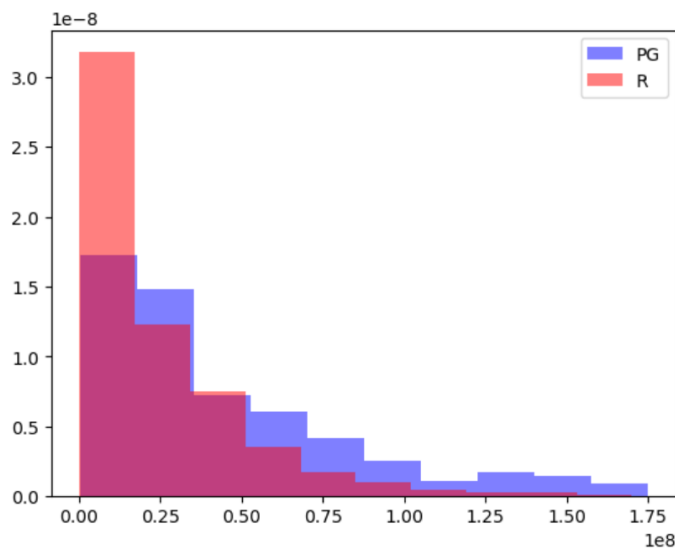
Based on the box plots, does movie rating appear to be associated with budget — that is, do PG and R movies have different budget distributions?

Skill 28.3 Exercise 1

Refer to the lists previously created in *Skill 26.1 Exercise 1*. One has the *production_budget* for PG movies and the other has the *production_budget* for R movies. Complete the spaces below to make overlapping histograms.

```
plt.hist(          , color="blue", label=          , density=          , alpha=          )  
plt.hist(          , color="red", label=          , density=          , alpha=          )  
plt.legend()  
plt.show()
```

Below are the actual overlapping histograms for PG and R movie production budgets,



Based on the overlapping histograms, does production budget appear to be associated with movie rating? Support your answer with evidence from the graph.

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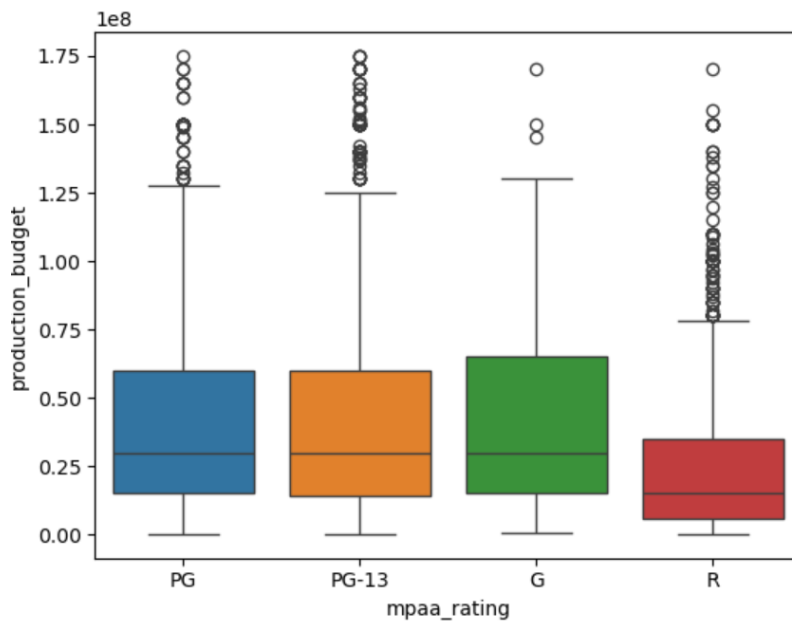
Skill 28.4 Exercise 1

Refer to the movies dataframe from above

Complete the spaces below to make side by side box plots for all *mpaa_ratings*

```
sns.boxplot(data = _____ ,  
            x = _____ ,  
            y = _____ ,  
            hue = _____ ,  
            legend = _____ )  
plt.show()
```

Below are the actual box plots for all *mpaa_ratings*



Based on the box plots, does movie rating appear to be associated with budget?